Progress Report: HPAI Risk Modeling in Canada using graph-neural network

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Progress Summary

My project aims to develop a reproducible machine learning pipeline for predicting poultry HPAI outbreak risk in Canada one month ahead. The broader modeling direction is to combine statistical and machine learning models with satellite-derived environmental data, poultry exposure data, road and transport features, and wild bird information. The final goal is to build and compare baseline models with graph-based and spatiotemporal graph neural network models.

At the beginning, we used 10 km grid cells as the spatial unit. However, for the graph neural network stage, I decided to move to Census Subdivisions (CSDs). This decision was made because CSDs provide an official and interpretable geographic structure, support privacy-safe aggregation of CFIA outbreak records, and allow different datasets to be linked to the same spatial unit. They also provide a practical unit for aggregating outbreak, poultry, environmental, and wild bird-related information when exact locations should not be used or displayed.

So far, I have built the static spatial foundation of the project. Each CSD is treated as one graph node. CFIA poultry outbreak records have been mapped to CSDs and converted into monthly outbreak labels. I also created the first set of static node features and built spatial adjacency edges between neighboring CSDs. This gives the project the basic graph structure needed before adding monthly dynamic predictors.

1. What is a Census Subdivision?

A Census Subdivision, or CSD, is an official Statistics Canada geographic unit. It usually represents a municipality, city, town, township, village, or similar local administrative area.

In this project, each CSD is represented as one graph node. This means the model can learn from the characteristics of each CSD and from its spatial relationship with nearby CSDs.

CSDs are useful for this project because they are:

- official and reproducible geographic units;
- easier to interpret than arbitrary grid cells;
- suitable for privacy-safe aggregation of outbreak records;
- linkable to agricultural, road, census, environmental, and satellite-derived datasets;
- useful as nodes in a national spatial graph.

The current graph contains **5,161 CSD nodes across Canada**.

2. Why Switch from 10 km Grid Cells to CSDs?

A 10 km grid cell has a fixed area of 100 km^2. CSDs do not have a fixed size; some are smaller than 10 km grid cells, while many rural or northern CSDs are much larger. Therefore, the reason for using CSDs is not that they are always spatially smaller. The main reason is that they are official, interpretable, and better suited for privacy-safe aggregation and graph construction.

Spatial Unit	**Strength**	**Limitation**
10 km grid cell	Fixed and spatially uniform resolution	Less directly connected to administrative reporting and privacy-safe aggregation
Census Subdivision	Official, interpretable, and useful for linking multiple datasets	Size varies across Canada

For this stage of the project, CSDs are more suitable because the graph neural network needs meaningful spatial nodes, and disease records can be aggregated without exposing exact private locations.

3. Summary of Work Completed

Component	**What I Completed**	**Why It Matters**
Spatial unit	Selected Census Subdivisions as the main unit for graph modeling	Provides an official, interpretable, and privacy-safe spatial structure
Graph nodes	Treated each CSD as one graph node	Creates the base structure needed for graph neural network modeling
Outbreak labels	Mapped CFIA poultry outbreak records to CSDs and converted them into monthly labels	Creates the target variable for one-month-ahead HPAI risk prediction
Static features	Built static CSD-level features from poultry, road, geography, and quality-control data	Describes baseline poultry exposure and spatial context
Graph edges	Created spatial adjacency edges between neighboring CSDs	Allows the future model to learn spatial relationships between areas
Dataset foundation	Combined nodes, labels, static features, and edges	Provides the core foundation before adding monthly dynamic predictors

4. Outbreak Label Construction

I mapped CFIA poultry outbreak records to their corresponding CSDs and converted them into monthly outbreak labels. This creates a privacy-safe CSD-month label table.

The label structure is:

Spatial Unit	Time Step	Label
One CSD	One month	Outbreak present or absent

This means the model does not need to use or display exact private outbreak coordinates. Instead, it learns whether a given CSD had a poultry HPAI outbreak in a given month.

The final prediction task is:

Input	Prediction Target
Current and past conditions for each CSD	Whether that CSD will have a poultry HPAI outbreak next month

5. Static Node Features Built So Far

I created the first set of static node features for each CSD. These features describe baseline conditions that do not change month by month, or that can be treated as relatively stable at this stage.

Feature Group	Examples	Purpose
Poultry production proxy	Statistics Canada poultry production and farm reporting data	Represents broader poultry exposure
Poultry density	GLW/FAO chicken and duck density layers	Captures spatial intensity of poultry presence
Road and transport exposure	Road density, major road density, highway density, distance to major roads	Represents movement and accessibility context
Geography	Area, province field, parent geography, spatial identifiers	Supports spatial organization and interpretation
Quality-control fields	Raster coverage flags, small-area flags, missing-data indicators	Helps identify unreliable or incomplete feature values

These features describe the baseline poultry exposure and spatial context of each CSD before adding monthly environmental predictors.

6. Spatial Graph Edges

I also created the first version of the spatial graph edges. These edges connect CSDs based on geographic relationships.

Edge Type	Meaning	Purpose
Rook edge	Two CSDs share a boundary segment	Represents strong geographic adjacency
Queen edge	Two CSDs touch at a corner or point	Represents weaker geographic adjacency

Nearest-neighbor fallback edge	An isolated CSD is connected to its nearest CSD	Prevents isolated nodes from being disconnected in the graph

These edges provide the first spatial backbone for future graph neural network modeling. In later stages, additional edge types can be added, such as environmental similarity, bird migration connections, transport similarity, or recent outbreak proximity.

7. Current Dataset Foundation

At this stage, the project now has the core dataset foundation needed for graph-based HPAI risk modeling.

Dataset Element	**Status**
CSD nodes	Completed
Monthly CSD outbreak labels	Completed
Static CSD node features	Completed
Spatial adjacency edges	Completed
Monthly dynamic predictors	Next step
Baseline models	Next step
Spatiotemporal GNN models	Future step

This is the base structure needed before adding dynamic monthly predictors.

8. Next Steps

The next step is to add monthly dynamic predictors because HPAI risk changes over time. These planned predictors include:

Dynamic Feature Group	**Examples**
Climate variables	Temperature, precipitation, humidity, snow, freeze-thaw conditions
Land cover and water features	Wetlands, cropland, surface water, seasonal water availability
Vegetation and moisture indices	NDVI, NDWI, and related satellite-derived indicators
Wild bird features	Monthly presence or abundance of relevant migratory waterfowl and reservoir species

After adding these monthly predictors, I will build temporal train, validation, and test splits. Then I will train simple baseline models first and compare them with graph-based and spatiotemporal GNN models.